

AI TRAINING · LG ELECTRONICS, NOIDA

Practical AI for the **Factory Floor**

A hands-on, 4-hour session - concepts, RAG, agents and orchestration, all no-code

Presented by **Arjun Thakur**

Principal AI Engineer · AI Trainer · Agentic AI Developer

FIRST - A FEW HANDS

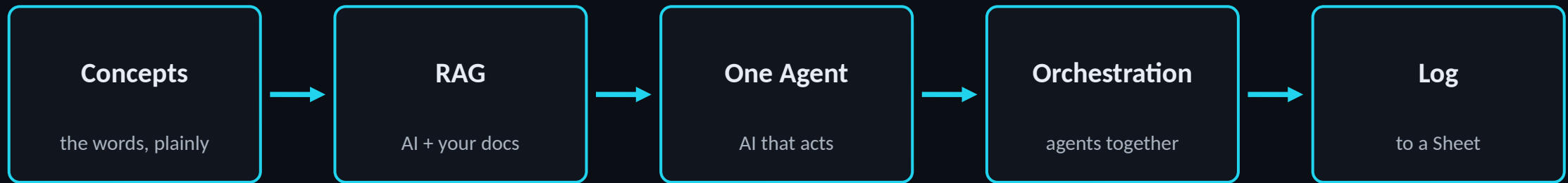
Before we start

- ▶ Who uses ChatGPT, Gemini or any AI chat - even once in a while?
- ▶ Who has ever written code or built an automation? (Excel formulas count!)
- ▶ Who has heard "AI agent" and wasn't quite sure what it means?

No wrong answers - this just helps me pitch the session at the right level for the room.

TODAY IN 4 HOURS

What we'll cover



- ▶ One example, grown all day - predictive maintenance on an IM-500 machine.
- ▶ I build live; you watch and ask. Short breaks along the way.
- ▶ You leave with a manual, prompts and a glossary to redo it yourself, free.

HOW WE'LL WORK TODAY

A few house rules

- ▶ Ask the moment you have a question - don't wait for me to finish. The best time is right away.
- ▶ Stuck on a word or a term? Ask me to define it, anytime. No question is too basic.
- ▶ Language is no barrier - Hindi or English, whatever's comfortable. Hinglish is perfectly fine.
- ▶ Everything is no-code and free. I build live; you get the manual to redo it yourself.

One example, grown all day

One Example, Grown All Day

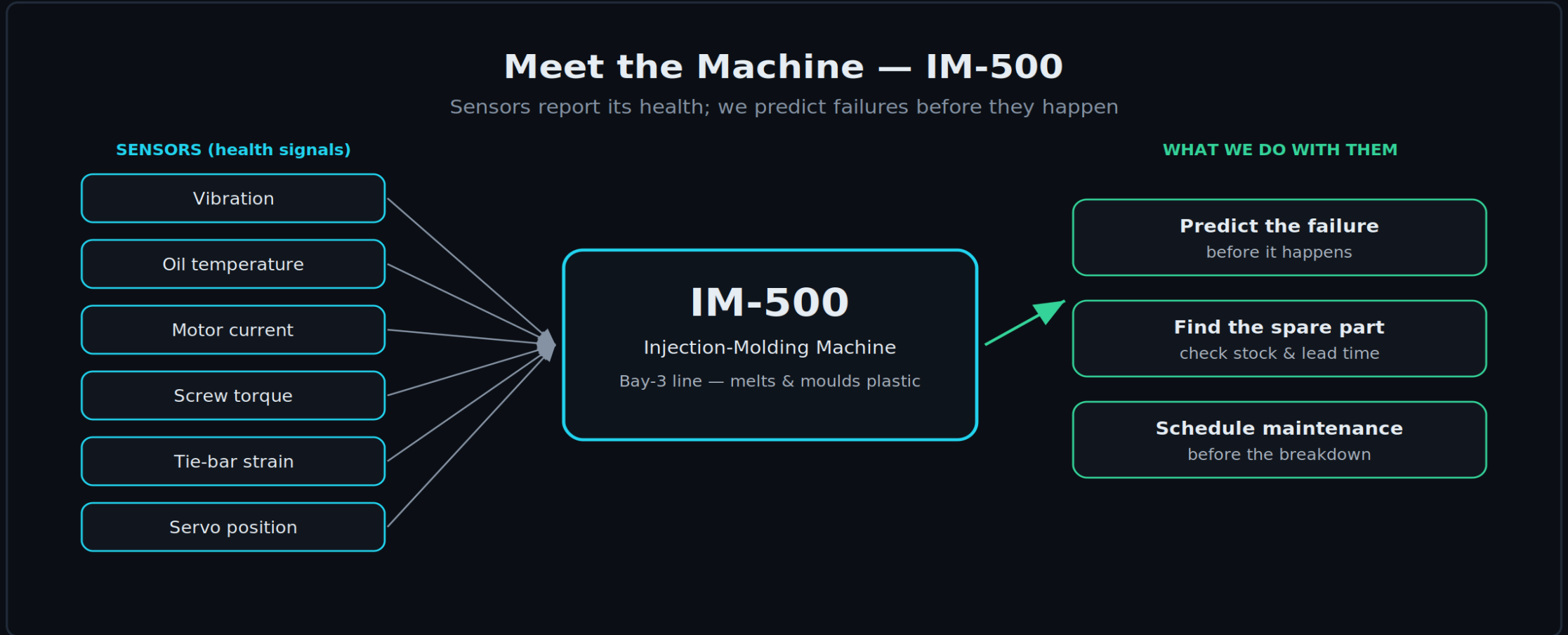
IM-500 predictive maintenance — the same story, deeper each stage



Same IM-500 example — it just gets deeper at every stage.

MEET THE MACHINE

The IM-500 - what we'll work on



01

SECTION 1

Concepts

The four words everyone mixes up - on the IM-500

AI vs GenAI vs AI Agent vs Agentic AI

AI vs GenAI vs AI Agent vs Agentic AI

Explained on the one machine we use all day — the IM-500

AI

Rules & patterns

"Flag if vibration > 4.5"

Generative AI

Creates language

Writes a shift summary from notes

AI Agent

LLM + your data + a decision

"Servo error -> EN-77 failing, ~6d" (Agent A)

Agentic AI

A team of agents coordinating

Predict -> parts -> maintenance (A,B,C)

Same IM-500 machine — four levels of capability. We climb this ladder today.

The four terms, defined

AI

- ▶ Software that follows rules or spots patterns
- ▶ No language, no reasoning - just "if this, then that"
- ▶ e.g. flag the part if vibration > 4.5

Generative AI

- ▶ Creates new content - text, summaries, images
- ▶ Understands and writes language
- ▶ Doesn't know YOUR data unless you give it (that's RAG)

AI Agent

- ▶ An LLM that can decide and take an action
- ▶ Uses your reference data (and tools) to finish a task
- ▶ e.g. reads a sensor alarm -> names the failing part

Agentic AI

- ▶ Several agents working together toward a goal
- ▶ They hand off, coordinate, span many steps
- ▶ e.g. predict -> check parts -> raise maintenance

Four terms, side by side

	What it is	Decides?	IM-500 example
AI	Rules & patterns	No	Flag if vibration > 4.5
Generative AI	Creates language	No	Writes a shift summary
AI Agent	LLM + data + decision	Yes, one task	Servo error -> EN-77, ~6d
Agentic AI	A team, many steps	Yes, end to end	Predict -> parts -> maintenance

Ask the room: *which of these are you already using without calling it "AI"?*

Q & A

Concepts - questions

- ▶ Which of your tasks today is really just automation?
- ▶ Where would a decision (an agent) actually help?

BREAK

7 minutes

We'll resume right on time.

02

SECTION 2

RAG

Making the AI answer from YOUR machine documents

The problem: a plain chatbot doesn't know your machine

- ▶ A raw AI model is smart, but it has never seen the IM-500 manuals.
- ▶ Ask it about servo error EN-77 and it guesses - a confident wrong answer.
- ▶ On the floor, a confident wrong answer can be worse than no answer.

What RAG is

How RAG Works — Retrieval Augmented Generation

Example: IM-500 predictive maintenance — sensor & spare-part lookup in NotebookLM

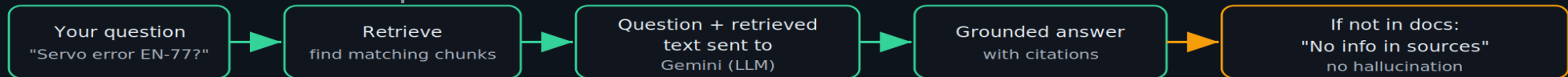
A. Prepare your documents (one time)



NotebookLM does all of this automatically when you upload files

index is searched at question time

B. Every time you ask a question



Why we use NotebookLM

- ▶ No-code, free, Google-stable - upload PDFs and ask; no database to manage.
- ▶ Best "just works" tool for grounded answers in a live demo.
- ▶ Reliability first: fewer moving parts, nothing to break on the day.

Before vs After RAG - the EN-77 question

Before RAG vs After RAG

Same servo-encoder question, with and without your documents

Same question: "Servo position error is rising — which spare part, is it in stock, what is the risk?"

BEFORE RAG

Plain chatbot, no company documents

Operator asks about the servo error + part

Model has never seen the IM-500 data

Result

- Generic or guessed answer
- Doesn't know the part number or stock
- May invent a lead time (hallucination)
- No source, not safe to plan on

AFTER RAG

Grounded in your uploaded documents

Same question about the servo error

Health Manual + Spare Parts + Addendum

Result

- Cause: servo encoder EN-77 degrading (~6 days)
- Exact part EN-77 — 0 in stock, 21-day lead
- Flags URGENT (lead time > failure date)
- Cited to the exact source — safe & actionable



Before vs After RAG, live

Leave the deck → [NotebookLM](#)

- ▶ Ask the EN-77 question in plain Gemini (no docs) -> vague guess.
- ▶ Upload the IM-500 manuals + spare-parts catalog -> ask again -> "not in sources" for the addendum item.
- ▶ Add the EN-77 addendum PDF -> ask again -> exact part, 0 stock, 21-day lead, cited.

[↗ notebooklm.google.com](https://notebooklm.google.com)

Q & A

RAG - questions

- ▶ Which documents would you point this at first?
- ▶ What shape is that data in today - PDFs, scans, spreadsheets?

BREAK

10 minutes

We'll resume right on time.

03

SECTION 3

Build an Agent

AI that reads, decides and takes action - no code

What makes it an agent

What Makes It an Agent

Agent A on the IM-500 — LLM + your data + a decision



Not just a chatbot: a brain (LLM) + your data + a decision. (Add a Tool and it can also act.)

Why we use n8n

- ▶ A no-code, visual builder with a free tier - drag boxes, connect them, done.
- ▶ Runs in a browser; no servers, no licences.
- ▶ You can rebuild everything yourself from the take-home manual.

Why Google Gemini - and a word on limits

- ▶ Free API tier, strong, instant key from Google AI Studio.
- ▶ Use a STABLE flash model (e.g. gemini-3.6-flash) - not a "preview" one.
- ▶ Preview models cap at ~20 requests/day; stable at ~1,500/day. Hit a limit? Switch to another stable model - the quota is per-model.



Build & run Agent A (Failure Prediction)

Leave the deck → [n8n](#)

- ▶ Chat Trigger -> AI Agent -> Google Gemini Chat Model -> paste the system prompt.
- ▶ Test: "vibration 3.9 mm/s for 4 shifts" -> pump bearing wear, ~7 days.
- ▶ Grounding check: ask something not in the data -> it refuses, no hallucination.

[↗](#) your n8n workflows home

Agent B - same pattern, different data

- ▶ Agent B = Spare Parts: given a failing part, returns stock + lead time + order decision.
- ▶ Identical steps to Agent A; only the reference data changes.
- ▶ Take-home: participants rebuild Agent B themselves from the manual.

Ask the room: *what's one lookup your team does by hand that an agent could do?*

Q & A

Agents - questions

- ▶ Which repetitive task could an agent take the first pass at?
- ▶ Where must a human always stay in the loop?

BREAK

7 minutes

We'll resume right on time.

04

SECTION 4

Agentic AI & Orchestration

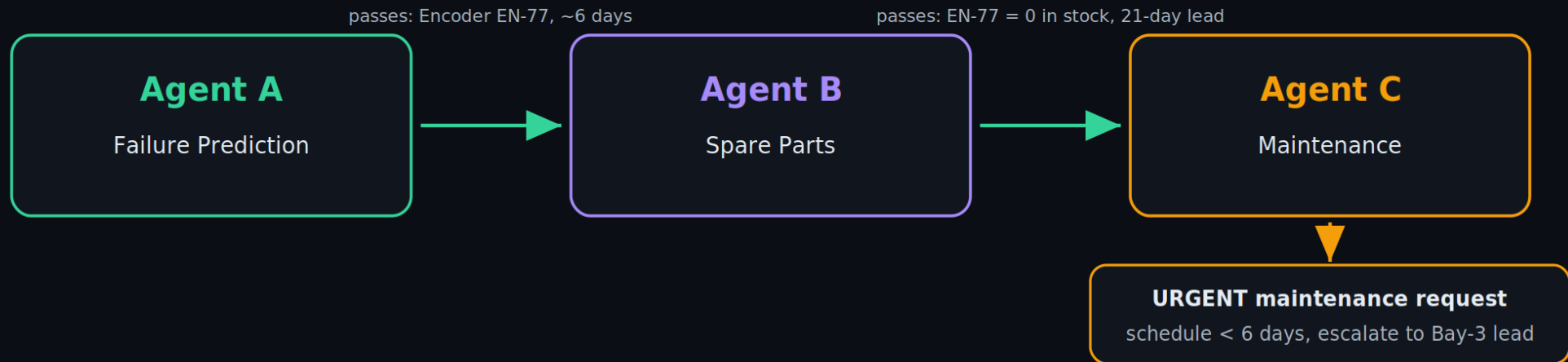
Many agents, working together

Agents as a team - A -> B -> C

Agents as a Team — Sequential Handoff

A -> B -> C: one question flows through all three

Input: "Servo position error 140 counts"



One question flows through all three — the final decision no single agent could make alone.

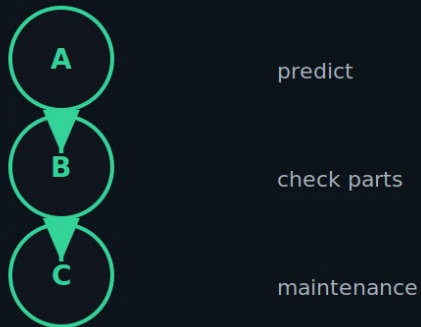
Three ways agents work together

Three Ways Agents Work Together

We use Sequential — simplest and most reliable

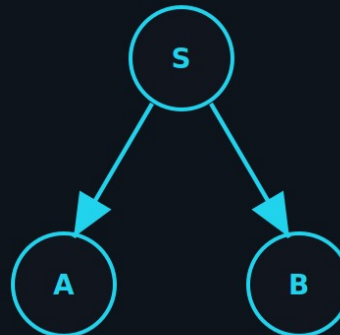
SEQUENTIAL

what we use — simplest, reliable



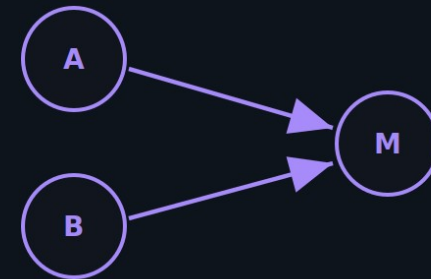
SUPERVISOR

one boss delegates



PARALLEL

run at the same time, then merge



All three are valid. For a clear, reliable live demo we build Sequential.



Wire A -> B -> C into one workflow

Leave the deck → [n8n](#)

- ▶ Copy the three agents onto one canvas; connect Chat -> A -> B -> C.
- ▶ Set the handoff prompts so each agent's output feeds the next.
- ▶ Ask ONE question - "servo error 140 counts" - and watch it flow to a final URGENT request.

Why this is powerful

- ▶ No single agent could produce the final decision.
- ▶ The escalation - 21-day part vs 6-day failure - emerges from the team, not one agent.
- ▶ This is "agentic AI": specialists coordinating toward a goal.

Ask the room: *where in your plant do several people already hand off like this?*

Why log to a Google Sheet (not email)

- ▶ A sheet is a permanent record you can filter, sort and share; email is a one-off notification.
- ▶ Native n8n node, same Google login - no extra setup.
- ▶ This is the agent taking a real action: it records, not just answers.



LIVE DEMO

Agent C writes the decision to a Google Sheet

Leave the deck → [n8n + Google Sheets](#)

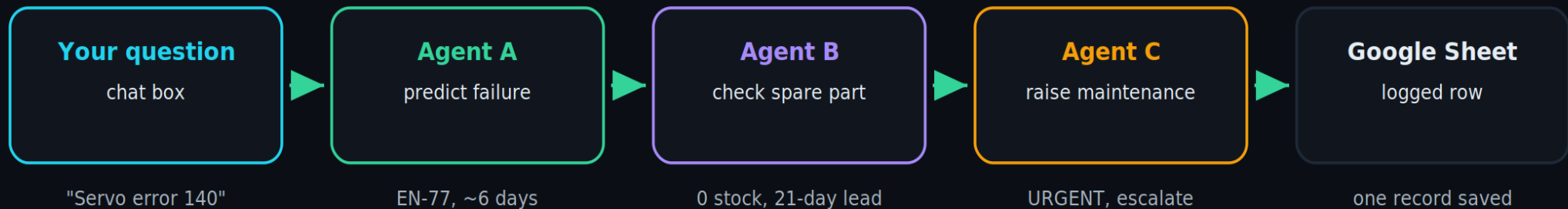
- ▶ Add a Google Sheets "Append Row" node after Agent C.
- ▶ Structured output maps cleanly into columns (component, part, priority...).
- ▶ Run the chat -> a new row appears in the sheet - the logged maintenance request.

[↗ sheets.google.com](https://sheets.google.com)

Question to logged decision

The Full Picture — Question to Logged Decision

IM-500 predictive maintenance, end to end



IM-500 predictive maintenance, end to end — the agent doesn't just answer, it records.

Q & A

Orchestration - questions

- ▶ Which two departments would benefit from talking to each other?
- ▶ Where would you want a human approval step?

What we covered today

- ▶ Concepts: automation vs AI vs agents vs agentic - in plain words, on one machine.
- ▶ RAG: making AI answer from your own documents, with citations.
- ▶ Agents: AI that reads, decides and acts - built no-code in n8n.
- ▶ Orchestration: three agents working together, logging a real decision.

WRAP - UP

Take it home

- ▶ You get: the how-to manual, the agent prompts, a glossary, the sheet template, and this deck.
- ▶ Rebuild it yourself - then swap in your own machine and data.
- ▶ Pick one small, real problem on your line to try first.

🔗 notebooklm.google.com · n8n.io · aistudio.google.com · sheets.google.com

Thank you

Questions any time - in Hindi or English.



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